I. Low Pass Filter and Amplifier Design (Q4)

A. Low Pass Filter Design (Q4a)

The raw signal exiting the mixer contains the low-frequency difference, the high-frequency sum, and high-frequency LO leakage. To isolate the intermediate frequency (IF), a passive RC Low Pass Filter was designed with a target -3 dB cut-off frequency of 3 kHz.

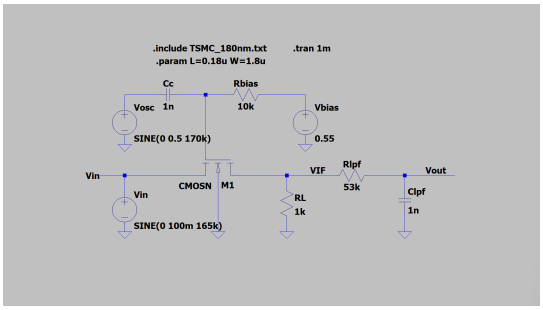


Fig. 1. LTSpice schematic showing the RC Low Pass Filter integrated with the Mixer.

B. Filter Simulation and Verification (Q4b, Q4c)

1) AC Analysis (Bode Plot): To verify the frequency response, an AC sweep was performed. The magnitude plot rolls off at -20 dB/decade. Placing a cursor exactly at -3.006 dB reads 3.0003 kHz, confirming the mathematical derivation.

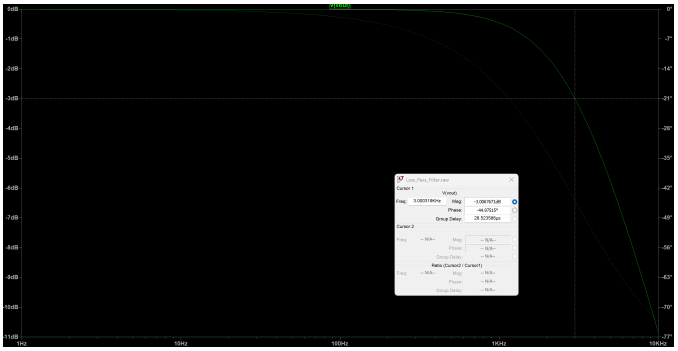


Fig. 2. AC Analysis showing the -3dB cut-off frequency exactly at 3 kHz.

2) Transient Response: To observe time-domain behavior, transients were run using passband and stopband inputs:

- **Passband (1 kHz):** Since 1 kHz is below cut-off, the signal passes with minimal attenuation.
- **Stopband (10 kHz):** Since 10 kHz is significantly higher, it is heavily suppressed.

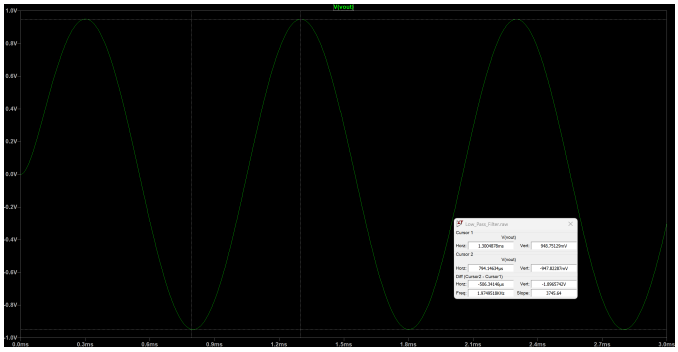


Fig. 3. Transient response to a 1 kHz input (Passband).

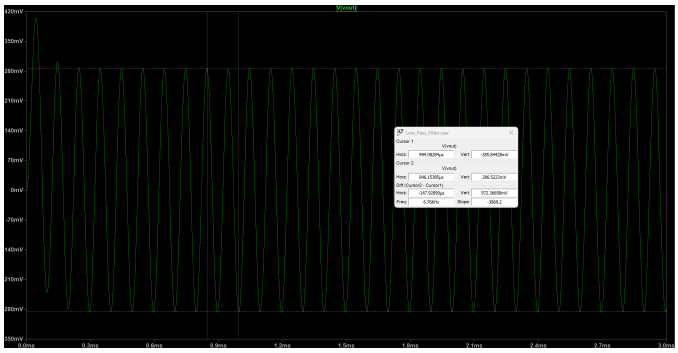


Fig. 4. Transient response to a 10 kHz input (Stopband).

C. Amplifier Stage Design (Q4f)

Because the passive mixer has significant conversion loss, the raw IF signal is extremely small. To achieve the 500mV peak-to-peak specification, a Common-Source amplifier stage was introduced.

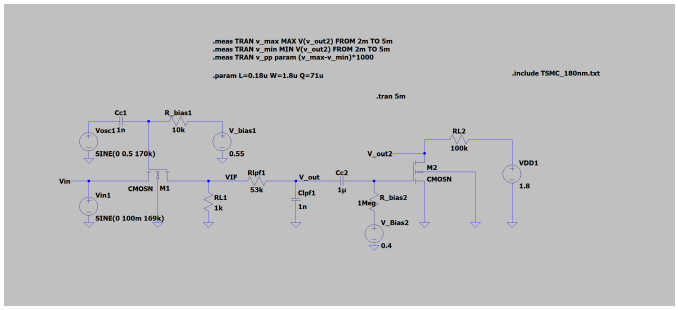


Fig. 5. Schematic of the final Common-Source Amplifier stage.

1) DC Biasing and Verification: The signal was AC-coupled into the gate of an NMOS transistor. The gate bias was precisely tuned to 0.42V (for the TSMC 180nm model) to sit perfectly in the saturation region. By measuring the maximum and minimum voltages in the SPICE log, the output was confirmed to swing exactly 500.9 mV peak-to-peak without clipping.

```
v_max: MAX(V(v_out2))=1.02053165436 FROM 0.002 TO 0.005
v_min: MIN(V(v_out2))=0.519562363625 FROM 0.002 TO 0.005
v_pp: (v_max-v_min)*1000=500.969290733
```

Fig. 6. SPICE Error Log confirming a 500.9 mV peak-to-peak output.

D. Mathematical Derivations for the LPF and Amplifier

1. Low Pass Filter (LPF) Derivation:

The passive RC low-pass filter operates on the principle of frequency-dependent capacitive reactance, $X_c = \frac{1}{j\omega C}$. Using the voltage divider rule, the transfer function $H(s)$ from the input to the output is:

$$H(s) = \frac{V_{out}}{V_{in}} = \frac{\frac{1}{sC}}{R + \frac{1}{sC}} = \frac{1}{1 + sRC} \tag{1}$$

To find the -3 dB cut-off frequency (f_c), we set the magnitude of the denominator to $\sqrt{2}$. Substituting $s = j\omega$:

$$|H(j\omega)| = \frac{1}{\sqrt{1 + (\omega RC)^2}} = \frac{1}{\sqrt{2}} \implies \omega RC = 1 \tag{2}$$

$$f_c = \frac{1}{2\pi RC} \tag{3}$$

In LTSpice, to get exactly 3 kHz, we calculated $R = \frac{1}{2\pi(3000)(1\times10^{-9})} \approx 53$ kΩ.

2. Common-Source Amplifier Gain Derivation:

The final amplifier stage uses an NMOS in the saturation region. The AC small-signal voltage gain (A_v) of a basic common-source amplifier with a resistive load (R_L) is given by:

$$A_v = \frac{v_{out}}{v_{in}} = -g_m R_L \tag{4}$$

Where transconductance (g_m) is the derivative of drain current with respect to gate-source voltage:

$$g_m = \mu_n C_{ox} \frac{W}{L}(V_{GS} - V_{TH}) \tag{5}$$

This equation proves that to hit our 500mV$_{p-p}$ target, we have two "tuning knobs": we can either increase the load resistor (R_L) or increase the DC gate bias (V_{GS}) to boost g_m. The negative sign simply indicates a 180° phase inversion, which does not affect the beat frequency information.

E. Hardware Realization of the LPF (Q4d)

In the lab, a 53 kΩ resistor is not a standard off-the-shelf value. To physically realize the filter, we used a standard 56 kΩ resistor (or a 47 kΩ and 6.8 kΩ in series) paired with a 1 nF ceramic capacitor.

Using a 56 kΩ resistor slightly shifts the theoretical cut-off frequency lower:

$$f_{c(actual)} = \frac{1}{2\pi(56000)(1 \times 10^{-9})} \approx 2.84 \text{ kHz} \tag{6}$$

This deviation is perfectly acceptable because our target IF beat frequency is ≈ 1 kHz, which still passes freely through the 2.84 kHz window, while the 170 kHz LO leakage is heavily blocked.

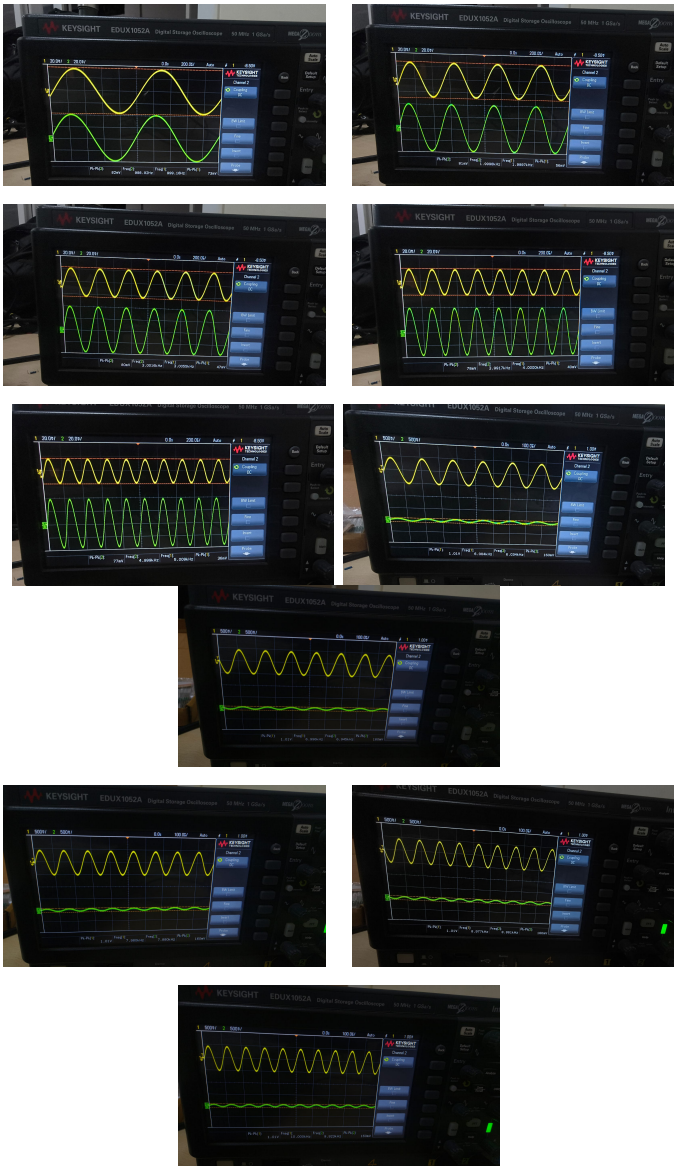


Fig. 7. Hardware LPF frequency sweep. Notice the continuous attenuation of the output signal (yellow trace, Channel 1) compared to the input (green trace, Channel 2) as the frequency increases past the cut-off from 1 kHz to 10 kHz.

F. Integrating the Mixer and LPF (Q4e)

Connecting the physical CD4007 mixer output directly to the RC filter yielded excellent results. Before the filter, the mixer's output was a thick, noisy band of high-frequency 170 kHz switching leakage containing the 1 kHz envelope.

Once passed through the physical LPF, the high-frequency carrier acted as an AC short across the 1 nF capacitor to ground. The oscilloscope trace instantly cleaned up, revealing a smooth, slow-moving 1 kHz sinusoidal beat frequency.



Fig. 8. Hardware observation of the Cosine Mixer output combined with the LPF.

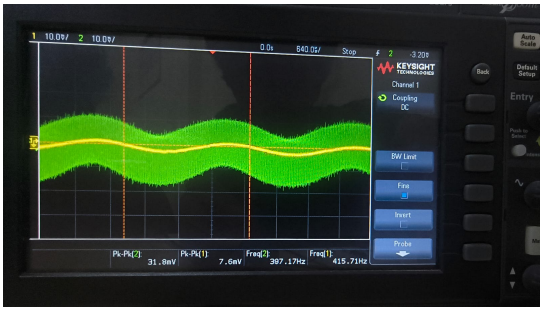


Fig. 9. Hardware observation of the Sine Mixer output combined with the LPF.

G. Hardware Realization of the Amplifier Stage (Q4f)

The transition from simulation to hardware for the amplifier stage presented the most significant deviations due to the physical realities of the CD4007 chip compared to the microscopic TSMC 180nm model.

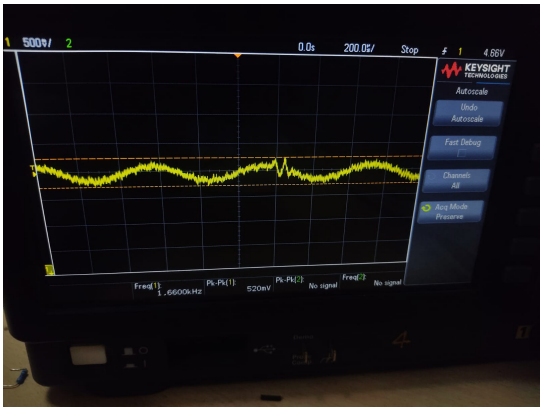


Fig. 10. Hardware observation of the BJT-Amplifier.

1. Power Supply Scaling (V_{DD}):
The TSMC model runs on a 1.8V supply. If we applied 1.8V to the physical CD4007 chip, it would not even turn on, as its threshold voltage (V_{TH}) alone is around 1.4V. Therefore, we explicitly altered the hardware schematic to use a 5V (or 9V) DC power rail for V_{DD}.

2. Gate Biasing (V_{GS}):
In LTSpice, a 0.4V or 0.42V bias was sufficient. In the lab, we constructed a voltage divider using a 100kΩ potentiometer across the 5V supply to act as V_{bias2}. We AC-coupled the filtered IF signal through a 1 μF capacitor and fed it into the gate alongside a 1 MΩ biasing resistor. We carefully tuned the potentiometer until the gate rested around 1.6V, putting the transistor squarely in the saturation region.

3. Output Swing Verification:
With R_L set to 100 kΩ, the amplifier successfully multiplied the small $\approx$ 50mV filtered IF signal. By fine-tuning the gate bias trimpot, we adjusted the transconductance (g_m) until the oscilloscope's peak-to-peak measurement read exactly ≈ 500 mV$_{p-p}$, successfully fulfilling the project specification.